VU2CPL Shack Changelog

Fixes and changes to non-DXCC tabs of the Node-RED shack automation (SPE amplifier, Power Control, Solar Conditions, Lightning, Rotator, etc.).

The DXCC Tracker has its own doc: see `README.md` / `DXCC_Tracker_README.pdf`.

2026-04-22

SPE Amplifier — Power meter scales with selected power level

Tab: SPE (`648eb83c2566c7b6`) **Node:** SPE → Dashboard Formatter (`8f7d96af1a4c24a9`)

The output-power bar was always scaled to a 1500 W full-scale regardless of the amp's selected power level. It now auto-scales based on `col10` (L / M / H):

Level	Full scale
L (Low)	500 W
M (Middle)	1000 W
H (Maximum)	1500 W

The bar text, percentage width, and amber/red thresholds (70 % / 90 %) all follow the new max automatically. Dashboard template (`e5b15d8e700bf002`) was already reading `d.pwrMax` — no UI change needed.

Power Control Panel — Fix stale "ON" defaults + reliable state refresh

Tab: All Power Strips (`b76a5310767803b4`)

Three related bugs resolved:

1. **HTML defaults lied.** Rotator, USB2, and USB3 plug tiles had `class="pw-plug on"` hardcoded in the template (`780b75182df31634`) — they showed ON on every page load until an MQTT `stat/.../POWER<n>` arrived. Since Tasmota only publishes `stat/.../POWER<n>` on a *state change*, the stale classes could persist indefinitely. All three now default to `off`.

2. **Dashboard ignored Tasmota's query responses.** `Power State → Dashboard` (`f8c3c072b381bd1c`) was filtering out `/RESULT` topics entirely, so responses to state queries (which Tasmota delivers on `stat/<device>/RESULT` as `{"POWER<n>":"..."}`) never reached the panel. The function now parses the RESULT JSON and extracts every `POWER<n>` key into the shared `power_states` object.

3. **Startup poll only queried POWER1.** The four `Poll <device>` function nodes (powerstrip1, powerstrip2, powerstrip3, 4relayboard) emitted a single empty-payload query to `cmnd/<device>/Power`, which Tasmota interprets as a POWER1 query only. They now loop over every outlet on the device, emitting one query per POWER:

```
for (var i = 1; i <= 5; i++) {
    node.send({ topic: 'cmnd/<device>/POWER' + i, payload: '' });
```

```
}
```

Combined with fix #2, this means every 30 s (and on deploy) the dashboard receives a fresh state for every outlet — the "always on" bug cannot persist more than ~30 s.

Solar Conditions — Geomagnetic K/A column indices

Tab: Solar (`590e889d44815afb`) **Node:** Parse K and A (`6e22622d54f653b1`)

The daily geomagnetic indices text file from NOAA (`services.swpc.noaa.gov/text/daily-geomagnetic-indices.txt`) has three index groups per data row:

Columns	Group
3	Middle-Latitude A
4–11	Middle-Latitude 8 × K
12	High-Latitude A
13–20	High-Latitude 8 × K
21	**Planetary A (Ap)**
22–29	**Planetary 8 × Kp (fractional)**

The parser was reading `cols[19]` for Ap and scanning `cols[20]..cols[27]` for Kp — that's the high-latitude K range plus the first six planetary Kp values, off by two columns in every case. Corrected to `cols[21]` for Ap and `cols[22]..cols[29]` for Kp. The length guard was also bumped from `< 28` to `< 30` .

Observed effect: yesterday's real values (2026-04-21) are A = 19, K ≈ 2.00. Before the fix the dashboard was showing A ≈ 3, K ≈ 4.00.

Solar Conditions — Aggregator branch discriminators

Tab: Solar (`590e889d44815afb`) **Node:** Solar State Aggregator (`d8ed9693432963df`)

The aggregator's SFI, K-Index, and A-Index branches all accepted any numeric `msg.payload` with a `msg.label` . In practice:

- The **SFI branch** matched every K / A message and happened to survive only because the SFI HTTP response arrived *last* in each poll cycle.
- The **K branch** had no discriminator against A, so the A-Index message (second output of Parse K and A) would overwrite `st.k` with the A value.

This was invisible while the parser was producing bogus near-identical K and A values (pre-column-fix). Once the column-index fix started producing a correct A = 19, the K gauge began showing 19.0 with the A's "Unsettled" label.

The three branches now discriminate by their gauge's `control.sectors` shape:

Metric	Discriminator

SFI	`sectors[0].val === 50` (gauge min is 50, not 0)
K	`sectors[0].val === 0.0 && sectors[1].val < 15` (K's 4.5)
A	`sectors[0].val === 0.0 && sectors[1].val > 10` (A's ≥ 35.5)

Rotator Auto-Off Timer — 60 s countdown reset loop

Tab: All Power Strips (`b76a5310767803b4`) **Node:** Rotator Auto-Off Timer (`05f0ddeb566a90fc`)

Symptom: rotator turned on, 60 s timer counted down to 0, then reset itself back to 60 s; physical switch stayed ON indefinitely. Cause: the timer unconditionally restarts whenever a `stat/powerstrip1/POWER2` = `"ON"` message arrives, and something (Tasmota echo, state-poll response, possibly a SetOption-related republish) keeps sending `"ON"` for an already-on outlet.

Fix: make the timer **idempotent** and add a post-OFF cooldown:

- If a timer is already running, additional `"ON"` messages are ignored (they do not reset the countdown).
- After the timer fires OFF (or an OFF is received), a 10 s cooldown begins. Any `"ON"` during that window is ignored — this breaks the reset loop regardless of its source.

Also fixed the status string from "Timer running — 5 min" to "Timer running — 60 s" (it has always been 60 s in code; the label was stale from an earlier intended value).

2026-05-01

AS3935 Lightning Sensor — Service revived after 10-day silent outage

Service: `as3935.service` (systemd) → `/home/vu2cpl/as3935_mqtt.py`

The lightning sensor daemon had been **dead since 2026-04-21 09:14 IST** and went unnoticed for 10 days because the dashboard had no liveness indicator. Discovered when a real storm fired no MQTT events.

Root cause of the original crash: at boot, `paho-mqtt`'s `client.connect()` called before the network was ready and threw `OSError: [Errno 101] Network is unreachable`. systemd hit `StartLimitBurst` and gave up. The script had no retry loop and no liveness signal, so silence looked identical to "no nearby strikes."

Hardening applied to `as3935_mqtt.py`

Change	Why
`mqtt_connect_with_retry()` (2 → 60 s exponential backoff)	Survives boot-time network race
Last Will & Testament on `lightning/as3935/status`	Broker auto-publishes `offline` if script dies
Startup retained publish to `lightning/as3935/status` (`event:"ready"`)	Visible in MQTT Explorer immediately
30-second heartbeat to `lightning/as3935/hb` (retained, includes `uptime_s` + per-event counters)	Silence > 60 s now unambiguously means dead

`on_connect` / `on_disconnect` callbacks log broker state	Easier diagnosis
I²C self-test on startup — reads `REG_CFG0` and logs the value	Catches dead-bus cases before the IRQ loop
Clean SIGTERM handler — publishes `offline` + GPIO cleanup	systemd `stop` no longer leaves a stuck GPIO claim
`try/except` around IRQ handler body	One bad I²C read can't crash the daemon
Event counters (lightning / disturber / noise / irq) in heartbeat	Observable from MQTT Explorer without reading logs

Antenna mode parametrization

The original script's `set_outdoor_mode()` wrote `0x0C` to `REG_CFG0`, which actually maps to `AFE_GB = 0x06` — *less* sensitive than the datasheet's "outdoor" value (`AFE_GB = 0x0E` → write `0x1C`). For the current indoor antenna placement, gain should be higher still: `AFE_GB = 0x12` → write `0x24`.

Replaced with `set_antenna_mode("indoor" | "outdoor")` controlled by a top-of-file `ANTENNA_LOCATION` constant. Currently `"indoor"` since the ferrite antenna is indoors. The mode is also published in the retained status payload so the dashboard can surface it.

Note: an indoor AS3935 has effective range of a few km at most (walls block the H-field) and is prone to false disturbers from local RF. Open-Meteo CAPE is the primary detection layer until the antenna is moved outdoors.

Required systemd unit override

`Environment=PYTHONUNBUFFERED=1` must be in `/etc/systemd/system/as3935.service.d/override.conf` — without it, `print()` output is block-buffered and never reaches `journalctl`. This is what made the running-but-silent state indistinguishable from "crashed."

Lightning Antenna Protector — Open-Meteo: `lightning_potential` is null in IN

Tab: Lightning Antenna Protector (`75e2cac8ab96f556`) **Nodes:** Build Open-Meteo URL (`3d61a4f561d40c70`), Parse Open-Meteo → Strike (`593f22a507b46335`)

Symptom: dashboard's OPEN-METEO MONITOR badge stuck at "Waiting for data…" indefinitely; auto-disconnect never fired even during real storms.

Root cause: the flow polled the `lightning_potential` parameter, which Open-Meteo's ICON-Global model returns as **all- `null` for India** (verified: 0/24 non-null entries for 13.065°N 77.806°E). Every poll hit `index === null`, the parser silently returned `null`, and nothing reached the dashboard or strike chain.

A second issue compounded it: the parser's mapping `km = (1 - index/100) × 200` assumed `lightning_potential` was a 0–100 percentage. Open-Meteo actually returns it as J/kg (LPI, an ICON-model quantity that typically ranges 0–25 even for severe storms). Even if the data had been present, the formula would have mapped a 25 J/kg severe storm to 150 km — well above any practical threshold — so disconnection would still never trigger.

Fix

Build Open-Meteo URL now requests `cape`, `weather_code`, and `precipitation_probability` (hourly + current weather_code). All three have continuous coverage globally.

Parse Open-Meteo → Strike rewritten with two outputs:

Decision	Synthetic km
`current.weather_code ∈ {95, 96, 99}` (TS now)	0 (overhead)
Hourly `weather_code ∈ {95, 96, 99}`	0
Showers (80–82) + CAPE ≥ 1000	20
CAPE ≥ 2500	10 (severe)
CAPE ≥ 1500	40 (strong)
CAPE ≥ 800	100 (moderate)
else	calm — no strike emitted

Output 1 → existing Parse Strike chain (only when storm-relevant). Output 2 → reserved for Master Dashboard live status (wire pending).

CAPE is *predictive* — it catches the convective setup hours before the storm peaks, which is what we want for a pre-emptive antenna disconnect. weather_code 95/96/99 is *deterministic* — when the model reports actual thunderstorm at the grid cell, force overhead.

Lightning Antenna Protector — Strike source label propagation

Tab: Lightning Antenna Protector (`75e2cac8ab96f556`) **Nodes:** Parse Strike (`26ddff0cbbfe5fc1`), Trigger Disconnect (`d62fb0c3c40f03b7`)

Event log lines for Open-Meteo strikes were appearing as `DISCONNECT: Blitzortung strike N km` because:

1. **Parse Strike Case 1** (object payloads from Parse Open-Meteo and test injects) constructed `msg.strike` without copying the upstream `payload.source` field. Source was lost on every object-shaped strike.
2. **Trigger Disconnect** had a fallback default of `'Blitzortung'` when `msg.strike.source` was missing — a leftover from when Blitzortung TCP was the original real-time source (Cases 2 and 3 in Parse Strike, which parse a binary/string TCP feed and are currently dead code — no upstream node feeds them).

Fix

```
// Parse Strike Case 1
msg.strike = { lat, lon, time, pol, region, source: d.source };  // added source

// Trigger Disconnect
const src = msg.strike && msg.strike.source ? msg.strike.source : 'unknown';
```

Cases 2/3 left intact — if Blitzortung TCP is wired in later, set `source: 'Blitzortung'` there at the same time.

Verified: log now reads `DISCONNECT: Open-Meteo strike 0 km`.

Lightning Antenna Protector — Dashboard refresh persistence

Tab: Lightning Antenna Protector (`75e2cac8ab96f556`) **Nodes:** Strike → Dashboard (`51367d456e71fb3e`), Master Dashboard (`557083037f168b22`), Refresh Stats (`61dca3d98a0e4c28`), Clear map every 30 min (`55f94d9dde0dc893`), new node Strikes Replay, new node Clear All

Symptom: refreshing the dashboard browser tab wiped the **event log** and **strike map markers** until the next event arrived. Both data sets were held only in dashboard-local JS variables (`var strikes=[]` and the `logLines` innerHTML), with no server-side persistence or replay path.

Fix — persist on the flow side, replay on the existing 30 s tick

1. **Strike → Dashboard** now appends each emitted strike to a flow-context array (`flow.strikes`), capped at 800 entries (FIFO). The `Format Log` node was already persisting log lines to `flow.event_log`.
2. **New `Strikes Replay` function node** — reads `flow.strikes` and emits `{type:'strikes_replay', list:[...]}` to Master Dashboard.
3. **`Refresh Stats`** (the existing 30 s ticker) now also fans out to:
   - `Log → Dashboard` (re-emits cached event log)
   - `Strikes Replay` (re-emits cached map markers)
4. **Master Dashboard** picks up a new message type:

```
if (d.type === 'strikes_replay') {
    strikeLayer.clearLayers();
    strikes = [];
    d.list.forEach(s => {
        strikes.push(s);
        L.circleMarker([s.lat, s.lon], {
            radius: 5, color: s.color, fillColor: s.color,
            fillOpacity: 0.9, weight: 1
        }).addTo(strikeLayer);
    });
    document.getElementById('lzCnt').textContent = 'Strikes: ' +
strikes.length;
    return;
}
```

5. **New `Clear All` function** sits between the 30-min clear-map inject and Master Dashboard:

```
flow.set('strikes', []);
flow.set('event_log', []);
return { payload: { type: 'clear' } };
```

Clears both the server cache and the dashboard simultaneously, so the next replay tick doesn't re-paint old strikes onto a freshly cleared map.

Behavior after fix

Action	Result
Page refresh	Log + strike markers re-appear within 30 s
New strike	Immediate update; also persisted to `flow.strikes`
30-min auto-clear	Map cleared on dashboard *and* server cache
Node-RED restart	Flow context lost (default). Add `flow.context.persistent=true` for cross-restart durability — separate change.

2026-05-06

AS3935 — IRQ pin was GPIO17 in software, GPIO4 in hardware

Script: `as3935_mqtt.py` **Helper:** `as3935_tune.py` (new)

Symptom after the May 1 revival: heartbeat counters stayed at `{lightning:0, disturber:0, noise:0, irq:0}` indefinitely. MQTT, I²C, and `status: "ready"` all worked, but no IRQ events ever fired.

Root cause

The original script declared `IRQ_PIN = 17`, but the AS3935 board's INT line was physically wired to **GPIO4** (Pi physical pin 7). Since the chip's I²C path was healthy, every diagnostic short of an end-to-end IRQ test passed. The mismatch had been there since first deploy; it was hidden because the chip-self-test only exercised I²C.

How it was found

A two-step diagnostic:

1. **Pull-test on GPIO17** — with the as3935 service stopped, sampled GPIO17 with `PUD_DOWN` / `PUD_OFF` / `PUD_UP`. Pin tracked the pull setting → genuine floating input → not connected to anything.
2. **SRCO scan** — set `DISP_SRCO=1` in chip register `0x08[6]` to route the chip's internal RC oscillator onto whatever pin its INT line is bonded to. Polled every accessible Pi GPIO (4, 5, 6, ..., 27. for transitions over 200 ms. **GPIO4 showed 12,162 transitions; every other pin showed 0.** Conclusive.

`IRQ_PIN = 4` change applied. Heartbeat counters started incrementing on the next NF=0 test (see below).

AS3935 — LC tank tuning to 500 kHz

Helper: `as3935_tune.py` (new)

The AS3935 has 16 internal tuning capacitor settings (0–15, ~8 pF per step) at register `0x08[3:0]`. The script wasn't using any of them — it relied on whatever stray capacitance the antenna+wiring happened to give. For accurate sferic detection, the tank should resonate at 500 kHz ± 3.5 %.

`as3935_tune.py`

New helper that:

1. Configures `LCO_FDIV = 3` (÷128 divider — slow enough for Python event-detection to count edges reliably; ÷16 at ~31 kHz overruns the user-space callback)
2. Sweeps `TUN_CAP` 0..15, samples 2 s of edges per setting, calculates frequency = `edges × 128 / 2 s`
3. Reports the table + recommends the cap value closest to 500 kHz
4. Flags out-of-spec results (>±3.5 %) with diagnostic hints

Must be run with `as3935.service` stopped (otherwise the daemon holds GPIO4) and the antenna in its **final mounting position** — moving the antenna afterwards changes stray capacitance and invalidates the tuning.

Result for current installation

TUN_CAP	pF	Hz	err %
0	0	516,480	+3.30
9	72	501,504	+0.30
10	**80**	**499,904**	**−0.02**
11	88	498,560	−0.29
15	120	493,056	−1.39

All 16 cap settings happened to be within spec (typical of a clean antenna circuit); `TUN_CAP=10` lands at −0.02 % err. Baked into the script as `TUN_CAP = 10`.

AS3935 — Add `CALIB_RCO`, INT flush, retained-status enrichment

Script: `as3935_mqtt.py`

Three additions to the chip-init sequence in the daemon, all chasing the same goal of making "is the chip happy?" answerable from MQTT without journal grep:

1. **`CALIB_RCO` at every startup** — the AS3935 has internal TRCO and SRCO oscillators that benefit from a one-time post-power-on calibration. The original script never called it; some chips work without, but timing-sensitive code paths can misbehave. Added: write `0x96` to register `0x3D`, wait 5 ms, verify `CALIB_DONE` (bit 7) set and `CALIB_NOK` (bit 6) clear in both `0x3A` (TRCO) and `0x3B` (SRCO). Result logged + published.
2. **Pending-INT flush after configuration writes** — register writes during init can transiently spike the IRQ line. Reading `0x03` (INT) clears it. Without this flush, the first real event sometimes arrived as a no-op rising edge that the `add_event_detect` callback then ignored because INT bits read 0.
3. **Retained `lightning/as3935/status` payload now includes** `tun_cap`, `irq_pin`, `calib_trco`, `calib_srco`. Status is re-published *after* calibration completes so the retained message reflects the actual result (the `on_connect` callback fires before init finishes).

Final init log

```
[init] AS3935 CFG0=0x24 (i2c bus 1 addr 0x03)
[mqtt] connected to 192.168.1.169:1883
```

```
[init] AFE_GB set for indoor antenna (CFG0=0x24)
[init] TUN_CAP set to 10 (80 pF)
[init] CALIB_RCO  TRCO=OK (0xA3)  SRCO=OK (0xA5)
[init] Cleared pending INT: 0x0
AS3935 ready — interrupt mode on GPIO4, noise_floor=4, antenna=indoor, tun_cap=10
```

Field verification

During the May 1 storm window (NF=0 momentarily), the chip caught **209 noise events** in roughly 30 s, then went silent as the storm moved off — the expected behaviour for an indoor narrow-band 500 kHz antenna. Subsequent tests with NF=2/NF=4 in clear conditions remained at 0 events: not a bug, just no nearby sferics for the selectively-tuned tank to resonate with. Until the antenna is moved outdoors, AS3935 is a confirmation-only secondary; Open-Meteo CAPE is the primary detection layer.

Lightning Antenna Protector — AS3935 dashboard liveness panel

Tab: Lightning Antenna Protector (`75e2cac8ab96f556`) **Nodes added:** AS3935 Status (mqtt in), AS3935 Heartbeat (mqtt in), Format AS3935 State (function), Replay AS3935 State (function) **Node modified:** Master Dashboard (`557083037f168b22`) — three handler inserts inside `scope.$watch('msg', …)`

Symptom: even with the AS3935 daemon healthy and publishing `lightning/as3935/status` (retained, on connect) and `lightning/as3935/hb` (retained, every 30 s), the dashboard's AS3935 panel header was permanently stuck at the static placeholder "Waiting…" because nothing in Node-RED forwarded those topics into Master Dashboard, and the panel only updated on event-driven `as3935_status` messages (which only fire when the chip detects something — never indoors with a 500 kHz selective antenna).

Fix — surface daemon liveness in the panel header

1. **Two new `mqtt in` nodes** subscribe to `lightning/as3935/status` and `lightning/as3935/hb` , datatype JSON, feeding a single `Format AS3935 State` function node.
2. **`Format AS3935 State`** caches each payload in flow context (`as3935_status` , `as3935_hb`) for refresh-replay and forwards typed payloads `{type: 'as3935_ready', …}` and `{type: 'as3935_hb', …}` to Master Dashboard.
3. **`Replay AS3935 State`** (new, fed by the existing 30 s `Refresh Stats` ticker) re-emits cached `flow.as3935_status` and `flow.as3935_hb` so a refreshed dashboard tab repopulates within 30 s — same pattern as the strike-replay fix from May 1.
4. **Master Dashboard** gained two new message handlers and a guard line on the existing `as3935_status` handler:
   - `as3935_ready` → sets the status LED green/amber/red based on calibration (TRCO + SRCO) and `event` field, writes `✓ READY · NF=4 · TUN=10` (or `⚠ CALIB?` / `OFFLINE`) to the header right-side text, and caches the last ready snapshot in `window._as3935Ready` for the heartbeat handler to render.
   - `as3935_hb` → updates the header text every 30 s with current uptime + IRQ count: `✓ READY · NF=4 · up 14m · irq=0` .
   - The existing `as3935_status` handler (disturber/noise) now sets `window._as3935EvtActive = true` for 60 s so a fresh event display isn't immediately overwritten by the next heartbeat; after the timeout it auto-reverts to ready/hb display.

Resulting panel states

Daemon state	Header text	LED

Up, calib OK, idle	`✓ READY · NF=4 · up 14m · irq=0`	🟢 green
Up, calib failed	`⚠ CALIB? · NF=4 · TUN=10`	🟡 amber
Offline (LWT)	`OFFLINE`	🔴 red
Disturber/noise event (60 s display)	`⚠ Disturber 23:18:01`	🟡 amber
Lightning strike	`⚡ N km (energy=…)`	red/amber/green by km
Page refresh while idle	re-populates within 30 s	🟢 green

Layout cleanup

While editing the dashboard template, the `<!-- Event Log -->` block was moved to sit immediately above `<!-- Map -->`. The log is more useful adjacent to the operator's eye-line for the lightning panel than buried at the bottom.

Standard Commit Sequence (reminder)

Per CLAUDE.md rule #4, extract the DXCC Tracker tab alongside flows.json:

```
cd ~/.node-red/projects/vu2cpl-shack
python3 -c 'import json; d=json.load(open("flows.json")); v=[n for n in d if
n.get("z")=="d110d176c0aad308" or n.get("id")=="d110d176c0aad308"];
json.dump(v,open("clublog_dxcc_tracker_v7.json","w"),indent=2)'
git add flows.json clublog_dxcc_tracker_v7.json
git commit -m "<description>"
git push
```

After push, on the Pi:

```
cd ~/.node-red/projects/vu2cpl-shack
git pull
sudo systemctl restart nodered
```